

CS 233 OBJECT ORIENTED PROGRAMMING AND DESIGN PATTERNS (OOPDP)



Shamama Anwar
Assistant Professor
Department of Computer Science and Engineering
Birla Institute of Technology, Mesra.
shamama@bitmesra.ac.in
9431103880

Module - II

Control Statements, Methods and Arrays

- Basic selection statements, Iterative constructs, Relative and Logical operators,
- break, continue,
- Methods, static methods, parameter passing, argument promotion and casting, scopes, method overloading.
- Arrays and ArrayList in Java, Enhanced for statement, Passing arrays to methods, Multidimensional arrays,
- Using command line arguments.

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Flow of Control

- Conditional Statements

- If – else (WAP to find the greater of 3 numbers)
- Switch – case (WAP to input a number between 1 – 7 and print the day of the week it corresponds to)

- Loops

- For (WAP to print all even numbers between 1 – 30)
- while (WAP to print all even numbers between 1 – 30)
- Do – while (WAP to print all even numbers between 1 – 30)

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Flow of Control

- break statement

```
import java.util.Scanner;
class demoContinue
{ public static void main(String args[])
{ int i;
  Scanner sc = new Scanner(System.in);
  double number, sum = 0.0;
  for(i=1;i<= 10;i++)
  { System.out.println("Enter a no.: ");
    number = sc.nextDouble();
    // If user enters negative number, loop is terminated
    if(number < 0.0)
    { break; }
    sum += number;
  }
  System.out.println("Sum :"+sum);
}
}
```

```
Enter a no.:
4
Enter a no.:
2
Enter a no.:
1
Enter a no.:
-2
Sum: 7.0
```

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Flow of Control

- continue statement

```
import java.util.Scanner;
class demoContinue
{ public static void main(String args[])
  { int i;
    Scanner sc = new Scanner(System.in);
    double number, sum = 0.0;
    for(i=1;i<= 10;i++)
    { System.out.println("Enter a no.: ");
      number = sc.nextDouble();
// If user enters negative number, loop is terminated
      if(number < 0.0)
      { continue; }
      sum += number;
    }
    System.out.println("Sum :"+sum);
  }
}
```

```
Enter a no.:
4
Enter a no.:
2
Enter a no.:
1
Enter a no.:
-2
Enter a no.:
:
:
:
Sum: .....
```

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Flow of Control

- Labelled break statement ([LAB Assignment](#))

```
public class BreakContinueWithLabel
{ public static void main(String args[])
  { first: for (int i = 0; i < 4; i++)
    { second: for (int j = 0; j < 4; j++)
      { third: for (int k = 0; k < 4; k++)
        { System.out.println ("inner start: i+j+k " + (i +
          j + k));
          if (i + j + k == 5)
            continue third;
          if (i + j + k == 7)
            continue second;
          if (i + j + k == 8)
            break second;
          if (i + j + k == 9)
            break first;
          System.out.println ("inner stop: i+j+k " + (i + j + k));
        } } } } }
```

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Flow of Control

- Write a program to check whether an input year is a leap year or not. (if)
- Write a program to ask the user to enter a number and print the number as Roman Numeral. (Switch-case)
- Write a program to print the sum of first 10 even numbers. (for, while, do-while)

Assignment Questions

- Given two int values, find their sum if the two values are the same otherwise print double their sum.
- Given an int n, print the absolute difference between n and 21, if n is less than 21 otherwise print double the absolute difference.
- Given 2 int, a and b, print true if one if them is 10 or if their sum is 10.
- Write a program to print Pascal's triangle. The number of rows are to be input by the user.

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Arrays

- Arrays can be declared using : `<type> [ ] variable_name;`
`int [ ] prime;      OR    int prime[ ];`
- No memory allocation at this point
- Arrays can be defined as: `variable_name=new <type>[N];`
`primes=new int[10];`
- Declaring and defining can be done in the same statement:
`int[] primes=new int[10];`

```
int k=7;
long[] primes = new long[k];
```

```
int k;
long[] primes =new long[k];
```

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Arrays

- When array is created, array elements are initialized
 - Numeric values (int, double, etc.) to 0
 - Boolean values to false
 - Char values to '\u0000' (unicode for blank character)
 - Class types to null
- JAVA checks whether the index values are valid at runtime
 - If index is negative or greater than the size of the array then an `IndexOutOfBoundsException` will be thrown
 - Program will normally be terminated unless handled in the `try {} catch {}`

```
long[] primes = new long[20];
primes[25]=33;
...
Runtime Error:
Exception in thread "main"
java.lang.ArrayIndexOutOfBoundsException: 25
at MorePrimes.main(MorePrimes.java:6)
```

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Arrays

- Array variable is separate from array itself
 - Like a variable can refer to different values at different points in the program array variables can also access different arrays

```
int[] primes=new int[10];
.....
primes=new int[50];
```

- Previous array will be discarded
- Cannot alter the type of array
- Array can be initialized as:

```
int[] even={2,4,6,8,10};
int[] value=even;
```

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Two Dimensional Arrays

- Two-Dimensional arrays can be created as:

```
float[][] temperature=new float[10][6];
```

- Can be initialized as:

```
int[][] array2D = { {99, 42, 74, 83, 100}, {90, 91, 72, 88, 95}, {88, 61, 74, 89, 96}, {61, 89, 82, 98, 93}, {93, 73, 75, 78, 99}};
```

```
float[][] samples;
samples=new float[6][];
samples[2]=new float[6];
samples[5]=new float[101];
```

```
int[][] ar = {{ 1, 9, 4 }, { 0, 2}, { 0, 1, 2, 3, 4 }};
```

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Two Dimensional Arrays

```
long[][] primes = new long[20][];
primes[2] = new long[30];
System.out.println(primes.length);
System.out.println(primes[2].length);
```

```
class unevenExample3
{ public static void main( String[] args )
{ int[][] uneven = {{1,9,4},{0,2},{0,1,2,3,4}};
  for ( int row=0; row < uneven.length; row++ )
  { System.out.print("Row " + row + ": ");
    for ( int col=0; col < uneven[row].length; col++ )
      System.out.print( uneven[row][col] + " ");
    System.out.println();
  }
}
```

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Call by Value vs. Call by Reference

- Primitive types are “call-by-value”
 - only a copy of the value is passed as an argument
 - method *cannot* change the value of the indexed variable
- class types are reference types (“call by reference”)
 - pass the address of the object
 - the corresponding parameter in the method definition becomes an alias of the object
 - the method has access to the actual object
 - so the method *can* change the value of the indexed variable if it is a class (and not a primitive) type

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Example: An Array as an Argument in a Method Call

```
public static void showArray(char[] a)
{
    int i;
    for(i = 0; i < a.length; i++)
        System.out.println(a[i]);
}
```

the method's argument
is the name of an array
of characters

```
-----
char[] grades = new char[45];
MyClass.showArray(grades);
```

uses the `length` attribute
to control the loop
allows different size arrays
and avoids index-out-of-
bounds exceptions

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Using = with Array Names

```
int[] a = new int[3];
int[] b = new int[3];
for(int i=0; i < a.length; i++)
    a[i] = i;
b = a;
System.out.println(a[2] + " " + b[2]);
a[2] = 10;
System.out.println(a[2] + " " + b[2]);
```

This does not create a copy of array a; it makes *b* another *name* for array *a*.

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Using == with Array Names

```
int i;
int[] a = new int[3];
int[] b = new int[3];
for(i=0; i < a.length; i++)
    a[i] = 0;
for(i=0; i < b.length; i++)
    b[i] = 0;
if(b == a)
    System.out.println("a equals b");
else
    System.out.println("a does not equal b");
```

tests if the *addresses* of *a* and *b* are equal, not if the array values are equal

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Methods that Return an Array

- The address of the array is passed
- The local array name within the method is just another name for the original array

```
public class returnArrayDemo
{ public static void main(String arg[])
  { char[] c;
    c = vowels();
    for(int i = 0; i < c.length; i++)
      System.out.println(c[i]);
  }
  public static char[] vowels()
  { char[] newArray = new char[5];
    newArray[0] = 'a';
    newArray[1] = 'e';
    newArray[2] = 'i';
    newArray[3] = 'o';
    newArray[4] = 'u';
    return newArray;
  }
}
```

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